

Experiments With a Michelson Interferometer

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Abstract

The aim of this experiment was to measure the wavelengths of the blue and yellow mercury lines to a high degree of accuracy (1%), as well as determining the calibration constant in the mechanical lever mechanism in micrometre. Additionally, the splitting of the yellow lines in mercury and sodium D was calculated. The experiment used a Michelson interferometer as well as a mercury and sodium light source. The calibration constant of the micrometer was calculated to be 3.604 ± 0.074 in arbitrary units. The wavelengths of yellow and blue light were determined to be 563.125 ± 14.001 nm and 493.698 ± 12.275 nm respectively. The separation of the two wavelengths in yellow mercury and sodium D light were 2.933 ± 0.434 nm and 0.600 ± 0.065 nm respectively. The main errors in this experiment were measurement errors associated with the accuracy of the micrometer and the accuracy of the calibration constant. In the line separation measurements, a systematic error was present due to the magnetic field of the earth.

1. Introduction

Interferometry as an experimental technique was introduced in the second half of the 19th century, pioneers such as Fizeau, Zehnder, Fabry and Michelson used the wave-like nature of light to study its properties, finding its speed and refractive index in different materials. Michelson used his version of an interferometer in his famous experiment to try and detect the aether. Interferometry continues to be used in a number of fields today, including astronomy, particle physics, spectroscopy and oceanography [1]. The most recent and perhaps the most famous experiment that utilised the interferometer developed by Michelson is the LIGO observatory in which scientists detected gravitational waves in 2017. Despite an obvious scale difference the LIGO experiment utilizes the same properties of electromagnetic waves as this experiment, performed in a university laboratory. Despite its small scale, the experiment demonstrates a few important insights into nature, for example, the separation of the Sodium D lines arise due to the Zeeman effect. Moreover, even with some inaccuracies, the experiment produces remarkably accurate wavelength results, with an uncertainty of less than one percent. [1]

2. Theory

The Michelson Interferometer superimposes two beams of light to observe interference. A schematic representation of the interferometer is shown in figure 1. Light from a light source is split in two by a beam splitter. The beams then move along the arms of the interferometer and are reflected by the two mirrors M1 and M2, recombining to produce an interference pattern observed at the detector.

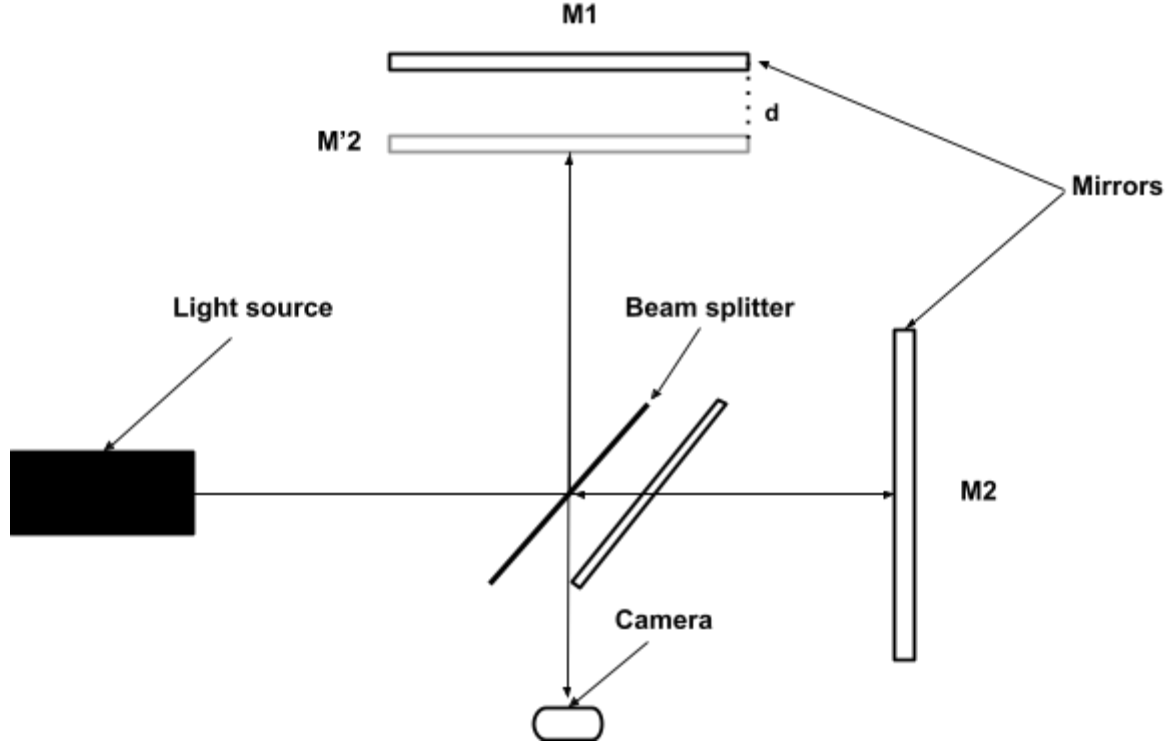


Fig. 1. Schematic representation of a Michelson interferometer, M1 is the movable mirror and both mirrors are tiltable. The black rectangle represents a 50/50 beam splitter. The clear rectangle represents a compensating plate with the same refractive index as the beam splitter and is included to equalize the optical path length along the arms of the interferometer.

M'2 is the location of the virtual image of the mirror M2 when viewed from the detector. The difference in path length between two on axis rays is therefore given by $2d$. However not all rays incident on the detector are parallel and may be an angle θ away from the axis. Constructive interference will occur when

$$2d \cos \theta = m\lambda \quad (1)$$

where m is an integer and λ is the wavelength of the light. For a constant d , m , λ and $\cos \theta$, the points of constructive and destructive interference (seen as dark fringes) will be spherically symmetric. Therefore by varying the position of mirror M1 by a known amount and counting fringes that pass by a reference point, the wavelength of the light source can be obtained using equation (2):

$$\Delta d = \frac{(\Delta n)\lambda}{2} \quad (2)$$

Where Δd is the change in displacement of M1 and Δn is the number of fringes passing a certain point.

Doublets

Two wavelengths λ_1 and λ_2 close to each will combine to form beats, points of high and low combined intensities. The superposition of the waves will vary between points of maximum and minimum intensity.

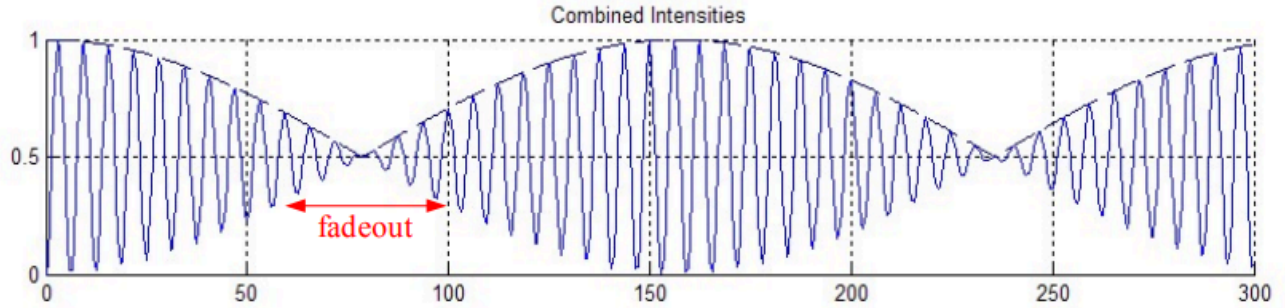


Figure 2. Diagram showing traveling wave packet composed of two waves.

The middle of each fadeout occurs when both waves are in anti-phase. This means that the two waves are a whole number of wavelengths apart, or that

$$2(d_2 - d_1) = n\lambda_1 = (n + 1)\lambda_2 \quad (3)$$

Where $(d_2 - d_1)$ is the path difference between the waves and n is a constant

Solving for n gives
$$n = \frac{\lambda_2}{\lambda_1 - \lambda_2} \quad (4)$$

Substituting back into equation 4 gives

$$\lambda_1 - \lambda_2 = \frac{\lambda_1 \lambda_2}{2(d_2 - d_1)} = \frac{\lambda_1 \lambda_2}{2\Delta d} \quad (5)$$

Using the average wavelength $\bar{\lambda}$, and defining, $\delta \equiv \bar{\lambda} - \lambda_1$ and $\epsilon \equiv \bar{\lambda} - \lambda_2$, means that

$$\begin{aligned} \lambda_1 \lambda_2 &= (\bar{\lambda} + \delta)(\bar{\lambda} + \epsilon) \\ &= \bar{\lambda}^2 + \delta\bar{\lambda} + \epsilon\bar{\lambda} + \delta\epsilon \\ &= \bar{\lambda}^2 - \delta^2 \\ &\approx \bar{\lambda}^2 \end{aligned} \quad (6)$$

Meaning that,

$$\lambda_1 - \lambda_2 = \frac{\lambda^2}{2\Delta d} \quad (7)$$

This gives a way of determining the wavelength separation given the average wavelength. [2]

3. Experimental details

3.1 Experimental Setup

The experimental setup is shown in figure 1. Two different light sources were used, a mercury lamp and a sodium lamp. The planes of the two mirrors were set to be perpendicular to one another by adjusting the tilt of mirror M2 so as to bring the two images of the light source into coincidence. Fringes were visible, and by small adjustments to the tilt, were expanded so that only a few fringes are visible in the field of view. The experiment consisted of four separate parts. In each part of the experiment a 16MP Samsung S5 phone camera was used to record a video of dark fringes passing a reference point. The video was then played at half speed to reduce random errors in counting the number of fringes.

3.2 Mirror Calibration

In the first part of the experiment the motion of the mirror M1 was calibrated by establishing a relationship between micrometer movement and the displacement of the mirror. A green filter was placed over a mercury light source and the number of fringes passing a reference point (Δn) was counted.

3.3 Wavelength of blue and yellow mercury lines

In the next part of the experiment the wavelength of blue and yellow mercury light was calculated using the calibration constant established in part 1. A blue and yellow filter were placed over the light source. Δn for a displacement of mirror M1 was found using the camera for both filters, equation 2 was used to calculate the wavelength.

3.4 Splitting of yellow mercury and sodium D lines

A third part of the experiment was centred around determining the splitting in the mercury yellow light and sodium D light. This was achieved by finding the distance between points of minimal intensity of the

combined beat. The micrometer screw was turned until the dark fringes became invisible. The screw continued to be turned until the next point at which the dark fringes became invisible. The distance between these points Δd was then used in Equation 7 along with the average wavelength of yellow light calculated in part 1 of the experiment to find the separation of the lines.

3.5 Errors

The main source of errors for the first two parts of the experiment was the precision of the micrometer which was 0.5 arbitrary units. Another source of errors was the random error in counting the number of fringes (fractional error of approximately 1%), however this was attempted to be reduced by the use of the camera. There was a systematic error throughout the experiment due to the uncertainty on the calibration constant; 0.074 in arbitrary units. Nevertheless, it accounts for less than 5% of the error value, therefore it is not significant. In the third and fourth part of the experiment, the significant sources of errors were once again random errors. However a major source of systematic errors, the ambiguity of determining when the points of minimal intensity in the beat become invisible, was unaccounted for. This may have distorted the wavelength difference values produced by the experiment. The simplest way to reduce the errors in the experiment would be to measure more fringes passing through the camera, as that would reduce relative error in the micrometer measurements of distance travelled. However, due to the high probability of human error while fringes were counted and time constraints we decided to stick with a maximum of 150 lines passing per measurement.

4. Results

4.1 Calibrating mirror motion

A green filter was placed over the light source, the wavelength of the green mercury line was taken to be 546.1nm. Three measurements were taken at different positions along the micrometer scale. For a micrometer change from 6.0 to 6.2, 117 fringes passed the reference point, for a change from 10.0 to 10.2, 132 fringes passed and for a change from 14.0 to 14.2, 149 fringes were measured. These results show that the relationship between micrometer movement and mirror displacement is not linear throughout the micrometer scale. To take into account any lack of linearity, all measurements were taken starting from a micrometer reading of 10. Using equation 2 and setting $d = kc$ where c is the distance moved on the

micrometer scale and k is a constant, k was found to be 3.6, meaning for a change of 0.01 units on the micrometer scale at 10, mirror M1 moves 3.6nm.

4.2 Measuring wavelength of mercury lines

The wavelengths of the blue and yellow mercury lines were measured using the calibration established in 4.1. Using a yellow filter placed over the mercury light source, a change in micrometer from 10.0 to 10.1 caused 64 fringes to pass the reference point. Using equation 2, the wavelength of the mercury yellow line was found to be 562.5 nm. Using a blue filter placed over a light source, a change in micrometer from 10.0 to 10.1 caused 73 fringes passed the reference point, giving a wavelength for the mercury blue line of 493.2 nm.

4.3 Separation of the mercury yellow line

The distance between points of minimal intensity fringes was found to be 0.015 on the micrometer scale, corresponding to a Δd value of 4 μm . Subbing this into equation 7 and using the average wavelength of yellow light found in section 4.2 yielded a separation of 2.93 nm.

4.4 Sodium D doublet

A sodium D light source was used for this part of the experiment. For a change in the micrometer from 10.0 to 10.2, 121 fringes were measured. Giving a wavelength for sodium of 595.0 nm. The distance between consecutive minima of fringe intensity was found to be 0.82 on the micrometer scale. Subbing this into equation 7 gave a value of 0.60nm for the separation, placing the wavelengths of the two sodium lines at 595.3 nm and 594.7 nm respectively.

5. Discussion and conclusion

Overall the wavelength values obtained for the blue and yellow mercury lines are consistent with tabulated values, within 3 standard deviations [3]. The largest fractional errors in the experiment were present in the mercury yellow and sodium D line splitting measurements, in both cases uncertainty exceeded 10% of the calculated value. This can be attributed to the systematic error in determining the exact point where the minimum intensity fringes become invisible. Another systematic source of errors that was not accounted for was the nonlinearity of the reduction factor. Even though the micrometer

readings were taken in relative proximity to the range at which the reduction factor was calculated, the differences between the reduction factor measured at the two ends of the micrometer scale were about 20%. In summary, the experiment successfully achieved the list of goals that were set at the beginning: wavelength values of blue and yellow light were obtained, a reduction factor of lever mechanism was calculated and size of Zeeman effect was calculated on a mercury yellow light and sodium D light.

References

1. Hariharan, P. (2007). *Basics of interferometry*. Amsterdam: Elsevier Academic Press.
2. Fowles, G. (2012). *Introduction to Modern Optics*. Dover Publications.
3. Bruno, T. and Svoronos, P. (2006). *CRC handbook of fundamental spectroscopic correlation charts*. Boca Raton, FL: CRC Press.